

MARY'S HARBOUR, NL, Canada

WMO: 713390

Lat: 52.304N

Lon: 55.833W

Elev: 38

StdP: 14.68

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.4	-9.8	-27.0	1.2	-12.4	-22.4	1.6	-8.5	36.3	24.8	32.7	23.4	8.8	250	0.819

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.9	78.5	63.6	74.6	62.2	70.7	60.4	65.7	75.6	63.6	72.0	61.6	69.2	14.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	83.1	69.9	59.8	76.8	67.7	57.8	71.4	65.8	30.6	75.8	29.0	71.8	27.5	69.4	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.0	26.4	23.2	-19.6	86.9	3.6	2.9	-22.2	89.0	-24.3	90.7	-26.4	92.3	-29.0	94.5		
(5)			-20.1	69.9	3.6	1.7	-22.7	71.1	-24.8	72.1	-26.9	73.1	-29.5	74.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	11.7	12.0	19.8	30.2	39.0	48.8	57.0	57.4	50.2	40.2	30.1	19.2
	DBStd	18.04	11.99	11.21	9.91	6.63	6.06	7.99	7.51	6.41	6.83	5.89	7.69	10.73
	HDD50	6219	1188	1063	935	595	349	120	21	8	78	309	599	954
	HDD65	11071	1653	1483	1400	1045	806	489	266	248	446	768	1048	1419
	CDD50	659	0	0	0	0	8	84	237	239	85	6	0	0
	CDD65	39	0	0	0	0	0	4	18	14	3	0	0	0
	CDH74	397	0	0	0	0	4	62	178	127	26	0	0	0
	CDH80	73	0	0	0	0	1	14	28	25	5	0	0	0
	WSAvg	10.7	11.5	11.7	12.3	11.6	10.0	10.0	8.9	8.7	10.0	10.6	11.3	11.9
	PrecAvg	42.5	4.4	3.7	3.2	2.4	2.8	3.0	3.5	3.7	3.7	3.9	4.1	4.3
(15) Precipitation	PrecMax	51.3	9.1	5.8	5.7	4.2	5.7	6.0	7.0	7.6	7.8	7.3	7.8	6.3
	PrecMin	35.2	1.8	1.9	1.3	0.8	0.7	1.2	1.4	1.1	1.3	1.6	1.7	2.0
	PrecStd	3.9	1.6	1.1	1.1	0.8	1.2	1.0	1.2	1.3	1.2	1.4	1.3	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	39.1	41.4	51.3	69.6	81.6	83.9	83.3	77.9	62.8	53.9	41.4
		MCWB	37.6	36.7	36.1	42.8	55.6	63.8	65.9	66.4	64.4	53.1	50.5	39.2
	2%	DB	35.5	34.6	36.9	45.6	60.6	73.8	78.8	77.2	69.7	57.4	47.8	37.7
		MCWB	33.9	33.3	33.5	39.6	50.2	59.8	63.6	63.8	59.7	51.2	45.3	36.4
	5%	DB	32.9	32.9	34.6	41.6	54.6	68.8	75.1	73.3	65.3	53.1	42.9	35.7
		MCWB	31.6	31.8	32.5	37.2	46.8	57.7	62.6	62.1	57.9	48.5	40.4	34.3
	10%	DB	30.4	30.4	33.0	38.6	49.1	63.3	70.8	69.3	61.3	49.3	39.6	33.7
		MCWB	29.1	29.2	31.3	35.0	43.5	55.0	61.1	60.5	55.0	45.4	37.7	32.3
	0.4%	WB	38.0	37.6	37.4	44.5	56.5	65.5	68.8	68.2	65.5	55.8	50.9	39.8
		MCDB	38.8	39.1	40.8	50.0	67.0	77.9	78.4	78.6	75.0	60.0	53.5	40.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.2	33.6	34.5	40.4	51.3	61.7	65.9	65.6	61.8	52.1	45.4	36.7
		MCDB	35.1	34.0	36.0	44.3	59.2	71.0	75.8	75.0	68.1	56.1	47.6	37.4
	5%	WB	31.9	31.9	33.0	37.7	47.1	58.6	63.6	63.4	58.7	48.9	40.8	34.8
		MCDB	32.9	32.9	34.2	41.0	53.7	66.8	72.6	71.1	64.3	52.5	42.3	35.6
	10%	WB	29.2	28.9	31.5	35.5	43.7	55.5	61.6	61.3	55.7	45.8	38.2	32.6
		MCDB	30.4	30.2	33.0	37.8	49.1	63.2	69.8	68.2	60.7	49.0	39.6	33.5
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	17.0	15.5	12.3	13.9	16.6	17.4	15.9	14.9	11.5	11.1	13.3
		MCDBR	15.3	14.6	14.6	16.6	27.1	27.8	25.9	24.1	21.9	18.9	14.2	11.4
		MCWBR	14.9	14.3	12.9	12.3	17.4	16.1	13.9	13.0	13.9	13.7	13.2	11.8
	5% WB	MCDBR	14.9	14.2	12.4	14.3	25.0	25.5	23.9	21.9	20.1	17.5	13.4	11.3
		MCWBR	14.8	14.2	11.7	11.2	16.9	16.1	14.2	13.2	14.1	13.8	13.3	11.6
(40) Clear-Sky Solar Irradiance	taub		0.274	0.284	0.302	0.357	0.392	0.385	0.411	0.381	0.353	0.321	0.292	0.282
	taud		2.293	2.281	2.272	2.228	2.276	2.375	2.347	2.433	2.480	2.533	2.477	2.292
	Ebn at Noon		231	265	283	278	272	276	266	268	264	252	228	204
	Edh at Noon		20	27	33	39	40	36	37	32	27	21	17	17
(44) All-Sky Solar Radiation	RadAvg		271	562	948	1357	1525	1580	1546	1303	971	560	298	190
	RadStd		40	66	149	143	179	152	142	85	69	49	29	25

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+3.53	N/A	N/A	+0.76	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	+2.57	N/A	N/A	+0.94	N/A	N/A	N/A	N/A

Nomenclature: See separate page